

LSV PA1 report

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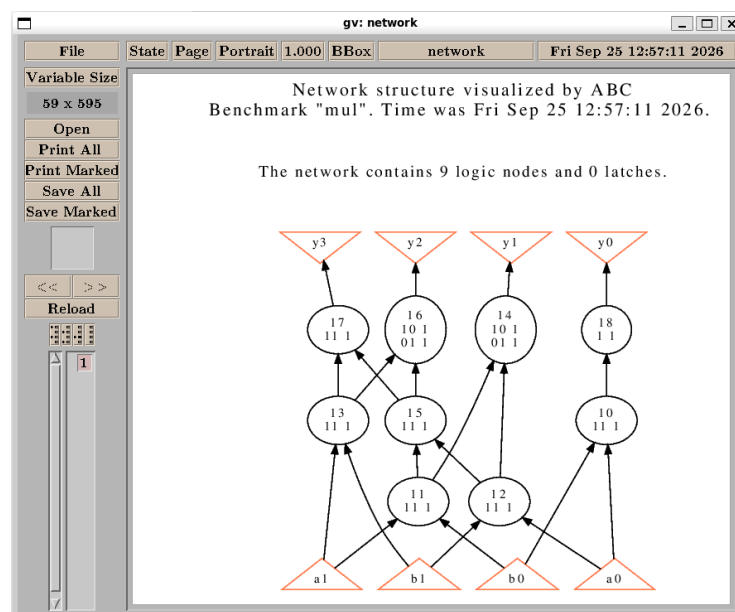
2.

(a)

```
.model mul
.inputs a1 a0 b1 b0
.outputs y3 y2 y1 y0
.names a0 b0 p00
11 1
.names a1 b0 p10
11 1
.names a0 b1 p01
11 1
.names a1 b1 p11
11 1
.names p10 p01 y1
10 1
01 1
.names p10 p01 c1
11 1
.names p11 c1 y2
10 1
01 1
.names p11 c1 y3
11 1
.names p00 y0
1 1
.end
```

(b)(1)(2)(3)

```
abc 01> read mul.blif
abc 02> print_stats
mul                                     : i/o =   4/   4 lat =   0 nd =   9 edge =   17 cube =   11 lev = 3
abc 02> show
```

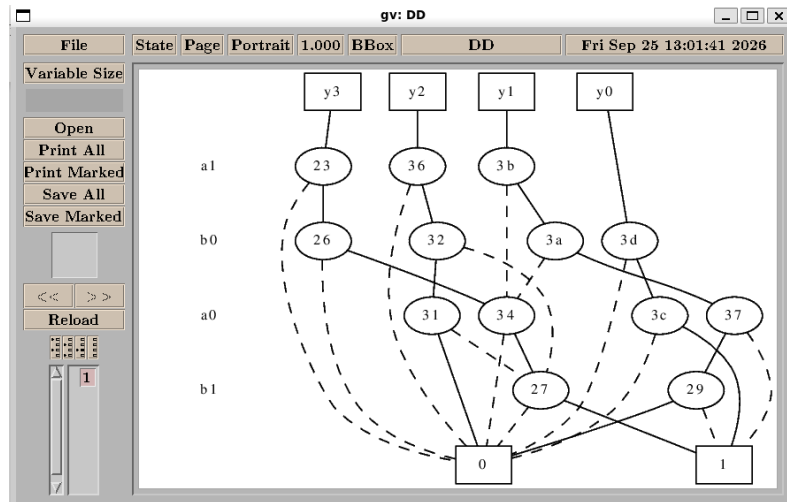


(b)(4)(5)

```

abc 03> collapse
abc 04> print_stats
mul
: i/o = 4/ 4 lat = 0 nd = 4 edge = 14 bdd = 14 lev = 1
abc 04> show_bdd -g

```

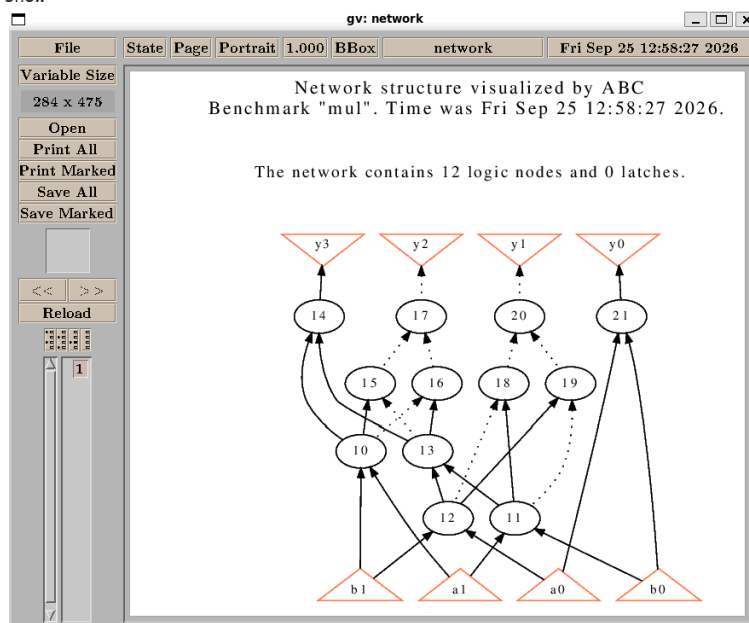


(b)(6)(7)

```

abc 02> strash
abc 03> print_stats
mul
: i/o = 4/ 4 lat = 0 and = 12 lev = 4
abc 03> show

```



3

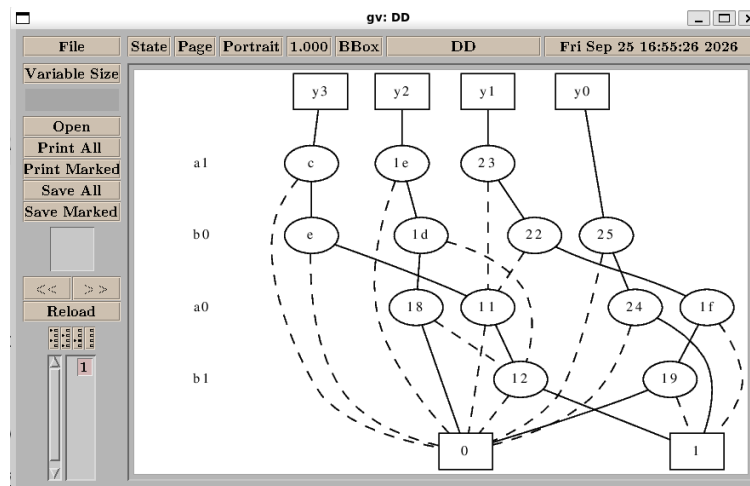
(a)(1)

The figures are 2(b)(3) and 2(b)(5)'s figures. They have the same AND nodes but with different number of levels.

(a)(2)

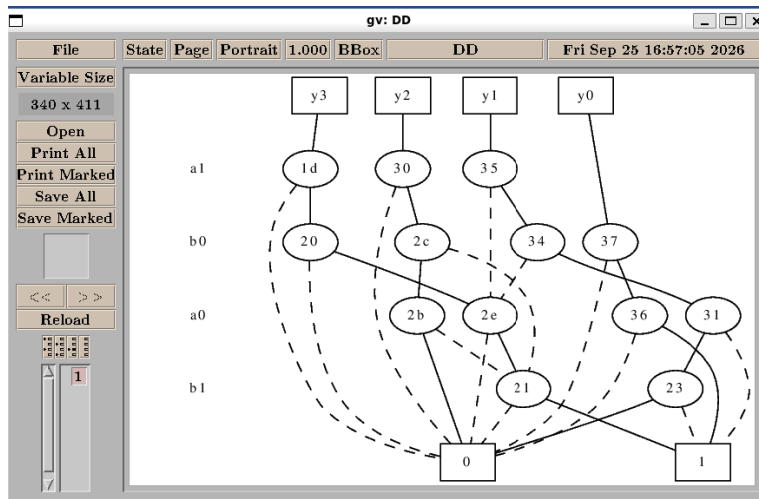
BDD:

```
abc 01> read mul.blif
abc 02> bdd
abc 02> print_stats
mul                               : i/o =   4/   4 lat =   0 nd =   9 edge =   17 bdd =   16 lev = 3
abc 02> show_bdd -g
```



Collapse:

```
abc 02> read mul.blif
abc 03> collapse
abc 04> print_stats
mul                               : i/o =   4/   4 lat =   0 nd =   4 edge =   14 bdd =   14 lev = 1
abc 04> show_bdd -g
```



After collapse, it has fewer edge, bdd nodes, and level.

(b)

```
abc 01> read mul.blif
abc 02> strash
abc 03> logic
abc 04> print_stats
mul                               : i/o =   4/   4 lat =   0 nd =  12 edge =   24 cube =   12 lev = 4
abc 04> lsv_print_nodes
```

```
Object Id = 9, name = n9
  Fanin-0: Id = 1, name = a1
  Fanin-1: Id = 3, name = b1
The SOP of this node:
11 1
Object Id = 10, name = n10
  Fanin-0: Id = 1, name = a1
  Fanin-1: Id = 4, name = b0
The SOP of this node:
11 1
Object Id = 11, name = n11
  Fanin-0: Id = 2, name = a0
  Fanin-1: Id = 3, name = b1
The SOP of this node:
11 1
Object Id = 12, name = n12
  Fanin-0: Id = 10, name = n10
  Fanin-1: Id = 11, name = n11
The SOP of this node:
11 1
Object Id = 13, name = n13
  Fanin-0: Id = 9, name = n9
  Fanin-1: Id = 12, name = n12
The SOP of this node:
11 1
Object Id = 14, name = n14
  Fanin-0: Id = 9, name = n9
  Fanin-1: Id = 12, name = n12
The SOP of this node:
10 1
Object Id = 15, name = n15
  Fanin-0: Id = 9, name = n9
  Fanin-1: Id = 12, name = n12
The SOP of this node:
01 1
Object Id = 16, name = n16
  Fanin-0: Id = 14, name = n14
  Fanin-1: Id = 15, name = n15
The SOP of this node:
00 0
Object Id = 20, name = n20
  Fanin-0: Id = 2, name = a0
  Fanin-1: Id = 4, name = b0
The SOP of this node:
11 1
```